Clinical XAI: A Tutorial on Concepts, Methods, and Modalities from Linear Models to LLMs

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Explainable AI (XAI) in clinical risk prediction helps make applied machine learning models transparent and clinically accountable. This tutorial covers recent progress across EHR, text, imaging, genomics, and multimodal data, with a later focus on emerging large language model (LLM) systems. We aimed to standardise terminology and distinguish terms such as interpretability, explanation, fairness, bias, trust, and transparency. We also provide a decision map linking methods to evaluation requirements. We review 97 unique studies (2019–2024) using a scheme that distinguishes alignment with knowledge (AK), expert review (ER), decision impact (DI), workflow integration (WI), and patient-facing (PF) assessment. We find that SHAP with tree ensembles is the most popular method in tabular EHRs, attention and CAM methods are common in imaging, and saliency-plus enrichment testing in genomics. A minority of papers report clinician-involved validation, and reporting of rater agreement, calibration, and subgroup effects is inconsistent. We, thus, propose a tiered, modality-agnostic framework, comprising AK+ER as the minimum standard, followed by DI (accuracy, time, confidence), and then WI (usage, outcomes), and provide design checklists, metrics, and pitfalls. Our aim is a roadmap, grounded in clinical practice, for developing, stress-testing, and deploying XAI in risk prediction.

CCS Concepts: • **Computing methodologies** → **Machine learning**; *Knowledge representation and reasoning*; • **Applied computing** → **Health informatics**; • **Human-centered computing** → Empirical studies in HCI.

Additional Key Words and Phrases: Explainable AI, clinical risk prediction, interpretability, healthcare AI, large language models, bias, fairness, clinical evaluation, electronic health records, medical imaging, genomics, multi-modal learning

1 Introduction

1.1 Motivation

Artificial intelligence can match clinicians in breast cancer screening, predict COVID-19 outcomes, and helps personalise patient care across all aspects of healthcare delivery [35, 80, 135]. But a core tension exists where models that deliver the best predictive performance are often the least explainable. In clinical settings, where decisions affect patient safety, resource allocation, and equity, explainable machine learning could reduce the chance of unsafe deployment. Explainability could help identify spurious correlations, incorrect error analysis, and detect the entrenchment of inequities by machine learning models, for example, where widely deployed risk scores have been shown to disadvantage Black patients relative to white patients [109, 154].

Trustworthy clinical AI, therefore, needs to be more than just highly accurate. Clinicians and patients need to understand how data are used and why recommendations are made [28]. Explainability supports appropriate reliance by helping clinicians detect model failures, identify bias, and communicate recommendations to patients [84]. With large language models (LLMs) and foundation models now being introduced to clinical workflows, the stakes are higher, both in potential harm and in usage potential. They inherit biases from their training data,

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can hallucinate plausible but false medical statements, and exhibit output variability, amplifying risks unless explanations and guardrails are part of the design [106].

Existing XAI methods, however, are heterogeneous and unevenly validated. Methods that work for images may not transfer to EHR time series, attention maps can be convenient "explanations" in large-scale deep learning models but their nature is non-causal, and additive attributions can fragment importance under correlation. Without some set of shared evaluation criteria, case studies cannot be compared. This tutorial aligns XAI methods to clinically meaningful evaluations (face validity, decision impact, workflow integration, and fairness), and shows how to report explanation quality alongside model performance. Our goal is to help researchers and practitioners build explanations that are safe, useful, and fair.

1.2 Contribution

This tutorial analyses explainable AI (XAI) methods for clinical risk prediction across multiple modalities, addressing reproducibility, validation, and evaluation. Unlike previous tutorials limited to some methods [23], we categorise the expanding XAI landscape, introduce LLM-specific challenges, and provide a characterisation of clinical evaluation methodologies in healthcare XAI. We examine each approach through the lens of practical deployment, asking:

(1) Does the model provide clinical benefit beyond existing practice?
(2) Could simpler, interpretable models achieve comparable performance?
(3) Have explanations been validated by clinicians in realistic settings?
(4) Has the explainability method undergone quantitative evaluation?
(5) How will benefits be communicated to patients?

We conclude with actionable recommendations for developing XAI systems that meet clinical practice requirements. Figure 1 presents our conceptual framework linking the terminology and methodologies covered throughout.

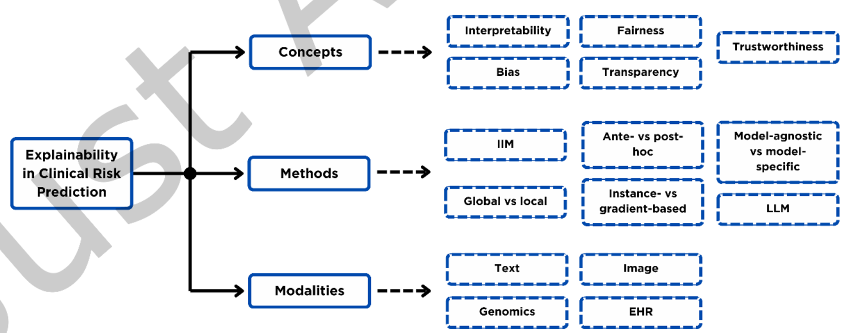

Fig. 1. Framework for the tutorial showing the key terms, methods, and modalities investigated under XAI for clinical risk prediction.

While raising these concerns, this tutorial is also designed to be a practical guide for researchers and practitioners implementing explainable AI in clinical risk prediction. We structure the content to progressively build competencies.

2 Preliminaries

2.1 Definitions and Key Terms

This section establishes conceptual boundaries between frequently conflated terms in explainable AI literature, necessary for evaluating clinical XAI systems.

Researchers often use explainability and interpretability interchangeably, but distinguishing them matters when reporting results. **Interpretability tells us how well the output predictions can be interpreted by behaviour in the input features**, for example, identifying which temporal patterns in vital signs impact acute risk predictions. Explainability includes interpretability but also includes functional understanding of algorithmic mechanisms, stakeholder interactions, and societal implications [138, 144]. Interpretation gives a functional mapping from input to output, but explanation requires understanding both the computation as well as its context, which could include bias detection, fairness, and stakeholder understanding [70].

Bias is systematic error that causes models to consistently make incorrect predictions for certain patient subgroups, especially when affecting vulnerable populations [43]. While some statistical bias is unavoidable, clinical AI has to separate benign pattern recognition and the perpetuation of healthcare inequities. We typically detect bias by analysing performance across demographic groups and use explainability to expose whether models rely on spurious correlations or genuine clinical factors [15]. A subtype of AI bias are cognitive biases in LLMs which represent systematic patterns of deviation from expected behaviour that mirror human cognitive biases [98]. Unlike demographic or representation biases, which come from training data imbalances, cognitive biases are predictable failures in response to input framing, anchoring information, or pattern frequency. **Fairness includes also the deployment of AI systems, addressing how biased predictions translate into differential treatment of vulnerable communities** [7, 136].

Researchers commonly assess fairness through counterfactual analysis and examine predictions under hypothetical demographic changes, stratified validation across subpopulations, and visualisation tools that reveal group-level disparities through techniques like FairSight and THEMIS [6, 20, 113]. These methods assume a priori knowledge of sensitive attributes, but struggle with intersectional vulnerabilities that do not fully fit with simple demographic categories.

Transparency implies the motivation, design, and implementation of an AI model are clear to stakeholders through documentation, testing, or other means, thus making it expressive enough to be human-understandable [49, 63]. Transparency requires that stakeholders understand system objectives, development choices, validation processes, and deployment constraints. Transparency in clinical AI requires early stakeholder engagement, iterative refinement based on clinical feedback, and documentation of limitations, without which one might have a complex and predictive AI that is clinically unsatisfying [34, 161].

Trust and accountability put these concepts into practice. Our framework builds trust through appropriate reliance metrics, pairing explanations with calibrated confidence scores and uncertainty quantification to prevent both over-reliance on incorrect predictions and under-utilisation of accurate ones [29, 52, 92].

Accountability includes answerability (who explains decisions?), liability (error responsibility), and auditability (retrospective review capability), often implying the need for clear documentation of limitations, audit trails linking predictions to model versions, override mechanisms, and error reporting processes [18, 167]. The EU's AI Act is a recent example of emerging regulatory frameworks mandating accountability infrastructure for medical devices, which could include high-risk clinical AI [198]. These boundaries between definitions help guide our evaluation of clinical XAI, where we assess whether published work has answered: Do authors identify important features and validate their clinical relevance? Has bias been evaluated across vulnerable populations? Were clinicians involved throughout development? Have explanations undergone robustness testing? As explanations do not imply performance guarantees, without rigorous validation or evaluation, they are approximations that may introduce additional error and misinterpretation without appropriate oversight [67, 70].

3 Taxonomy of Methods

We organise clinical XAI along four practical design axes: (1) *inherent vs. post-hoc*; (2) *local vs. global*; (3) *model-agnostic vs. model-specific*; and (4) *attribution vs. concept/prototype vs. counterfactual*. This aligns with prior healthcare-focussed surveys while keeping risk prediction as the primary use case [22, 157]. In this section, we provide a tutorial-style taxonomy of explainability methods for clinical risk prediction. We synthesise recent comprehensive surveys, including the PRISMA-based review by Hossain et al. [84], the taxonomy by Sun et al. [188], and the survey by Bilal et al. [25], with practical guidance for implementation and evaluation in clinical settings, we also connect these to earlier healthcare and domain-general surveys.

3.1 Recent Surveys

Recent surveys and viewpoints motivate several, partly overlapping, ways to categorise XAI in medicine by:

(1) **Explanation mechanism** attribution (what contributed) versus non-attribution (why/what-if via concepts, prototypes, counterfactuals) [15, 78, 84, 188].
(2) **Transparency** inherently interpretable (glass-box) versus post-hoc (explaining opaque models) [22, 67, 84].
(3) **Scope** local (individual-level) versus global (model-level) explanations [78, 157, 188].
(4) **Implementation family** gradient/back-prop rules, perturbation/surrogate, Shapley-based, concept/prototype, and counterfactual methods [15, 84, 188].

As shown in our framework (Figure 1), these dimensions interact; e.g., an EHR model may couple a model-agnostic local explainer (TreeSHAP) with attention heatmaps for temporal saliency [84, 157].

Rather than re-survey XAI at large, we provide a practical, risk-prediction–centred tutorial that: (i) organises methods by the four axes above; (ii) spans all major data modalities for clinical risk prediction (EHR tabular/time-series, text, imaging, and genomics); and (iii) foregrounds evaluation and clinical validation (what to measure, how to validate with clinicians, and typical failure modes). To make this concrete, Table 1 catalogues representative methods per family with scope, model-specificity, and examples (including 22, 25, 67, 84, 157, 188) that readers can follow up in depth.

3.2 Inherently Interpretable Models (IIM)

IIMs or, as they are often termed, glass-box or white-box models due to their relative decision-making transparency are among the simplest methods for explainable prediction in clinical risk modelling. Their transparency is due to the simplicity of the internal structure of the model and the relationship between input and output that can be clearly represented either through equations or visualisations. They do not need an external model or method to make the predictions explainable [171]. Many such methods predate machine learning itself, with linear regression (least squares) being an example.

Implementing Sparse Linear Models. One of the most common groups of methods here is sparse (linear) classifiers. They include a collection of models from linear and logistic regressions to wider generalised additive models. In linear or logistic regression models, the output is modelled as a linear combination of the features or assuming a Gaussian distribution for the outcome conditional distribution, thus making assumptions that might be violated in a non-linear or non-Gaussian setting. To overcome this, generalised linear models (GLMs) assume a non-Gaussian conditional distribution for the outcome, relating its mean to the same linear combination of features through an invertible link function [196]. A general equation subsuming this family of methods can be written as:

$$g\left(E_Y(y \mid x)\right) = \beta_0 + f_1\left(x_1\right) + f_2\left(x_2\right) + \ldots + f_p\left(x_p\right) \tag{1}$$

g is a link function that maps the conditional expectation to the linear predictor scale, accommodating both classification and regression settings. $E_Y(y \mid x)$ is the expected outcome, assumed Gaussian in linear regression

Table 1. Catalogue of explainability method families for clinical risk prediction (representative, not exhaustive). Each row lists exemplar techniques, typical scope (the natural unit a method addresses, not whether results can be later averaged), whether model-agnostic (MA) or model-specific (MS), and anchors for further reading.

Family	Exemplar techniques	Scope	MA/MS	Key refs
Inherently interpretable	GLMs; GAM/EBM; sparse scoring (SLIM); decision/rule lists; Cox	Global	MA	[22, 30, 193]
Attribution (gradient & CAM)	Integrated Gradients; DeepLIFT; LRP; Grad-CAM/++; SmoothGrad	Local	MS	[143, 177, 181, 183]
Attribution (perturbation & surrogates)	Occlusion; Meaningful Perturbations; LIME; Anchor	Local	MA	[64, 170]
Shapley-based	SHAP (Kernel/Tree/Deep); Expected Gradients	Local & Global	MA/MS	[19, 125]
Prototype / influence	MMD-Critic; Influence Functions; ProtoDash; ProtoPNet	Local & Global	MA/MS	[38, 79, 107]
Concept-based	TCAV; ACE; ConceptSHAP; Concept bottlenecks	Global for TCAV/ACE; ConceptSHAP. Local for CB	MS	[69, 105, 108, 178]
Counterfactual / recourse	Wachter CF; DiCE/FACE/MACE; task-specific recourse	Local	MA	[200, 205]
Attention (sequences & vision)	Additive/self-attention; Transformers (attention rollout)	Local & Global	MS	[37, 197]
Time-series (EHR)	SHAP on GBDTs; temporal attention maps; TCN saliency	Local & Global	MA/MS	[55, 157]

Abbreviations: MA—model-agnostic; MS—model-specific; EBM—Explainable Boosting Machine; LRP—Layer-wise Relevance Propagation; TCN—Temporal Convolutional Network.

or drawn from another exponential-family distribution in a GLM. β_0 is the intercept, while the component functions f_j reduce to linear coefficients $\beta_j x_j$ in standard linear and generalised linear models, or may be learned as arbitrary smooth functions in generalised additive models (GAMs). Extracting the fitted weights gives a direct, quantitative interpretation of each feature's contribution [196]. The choice of link function can limit interpretability by breaking the simple relationship between the weighted sum of inputs and the output.

Practical Tip: When implementing GLMs for clinical risk prediction, one should start with simple link functions (logit for binary outcomes, log for count data) before exploring more complex transformations.

Before moving on to discretisation methods that address some of the limitations of linear models, an important aspect of sparse linear models is their sparsity. Using a large number of features makes prediction problems often less tractable and the models prone to overfitting (even when regularisation methods are applied), and having to interpret too many features reduces readability of the explained results [217]. Supersparse linear integer models (SLIMs) have been developed to address these challenges and tested on medical applications. SLIMs produce integer-valued risk scores using fewer features than regularised linear models. They use a linear form that helps users to gauge the influence of each input variable with respect to the others while encoding interpretability requirements through a rule-based format [193]. This is done through a balanced loss between higher prediction accuracy and higher sparsity while using restricted and discrete coefficients $\boldsymbol{\lambda}$ for sample $\boldsymbol{x}_i$ out of N total samples:

$$\begin{array}{ll} \min_{\lambda} & \frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\left[y_i\boldsymbol{\lambda}^T\boldsymbol{x}_i \leq 0\right] + C_0\|\boldsymbol{\lambda}\|_0 + \epsilon\|\boldsymbol{\lambda}\|_1 \\ \text{s.t.} & \boldsymbol{\lambda} \in \mathcal{L}. \end{array} \tag{2}$$

SLIM minimises the $0-1$ loss $\frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\left[y_i\boldsymbol{\lambda}^T\boldsymbol{x}_i \leq 0\right]$ and ℓ_0-norm $\|\boldsymbol{\lambda}\|_0 := \sum_{j=1}^{P}\mathbb{1}\left[\lambda_j \neq 0\right]$ with penalty C_0 for norm and P features. Coefficients are restricted to, for example, $\mathcal{L} = \{-10, \ldots, 10\}^{P+1}$, and can include operational constraints like $\|\boldsymbol{\lambda}\|_0 \leq 10$. SLIM uses a regularisation ℓ_1 factor $\epsilon\|\boldsymbol{\lambda}\|_1$ in the objective to restrict coefficients to coprime values. The parameter ϵ is set to a negligible value to avoid ℓ_1-regularisation effects [193].

Worked Python implementations of SLIM and the Explainable Boosting Machine, with a clinical deployment checklist, are provided in Listing S1 of the supplementary material.

Discretisation Methods. Discretisation methods reduce a problem and its data into a set of discrete subsets to transform the continuous solution space into discrete points. Decision trees and their ensembles learn by evaluating each feature value and selecting the cut-off to maximize the class separation as measured through entropy or Gini impurity (classification) or minimise the variance of the outcome (regression). Thus, each feature's importance can be interpreted as share of the overall model performance [140]. A limitation is that in cases of linear relationships between input and output, it approximates the behaviour through a series of discrete step functions which might not be of high fidelity.

Other discretisation methods include rule-based or expert-based systems, one of the earliest examples of domain adaptation of machine learning to healthcare practice [134]. Like decision trees, these methods decompose outcome modelling into discrete IF-THEN steps whose cut-off criteria are not determined by optimising a certain metric but by domain expertise or, in the case of Bayesian rule lists, from data mining the features for patterns [118]. As long as the decision rules are understandable, these models are radically interpretable, simple, and often more compact than decision trees. Through domain expertise, they also apply a certain type of feature selection by *a priori* defining what features are of importance from expert knowledge and only including those in the decision rule structure whereas nonsparse linear models would assign weights to all features and learn a less sparse model. Similarly to decision trees, they are not well suited to linear modelling problems while, additionally, only being able to solve classification problems with discretised features.

Instance-Based Interpretability. Unlike the parametric models discussed above, k-Nearest Neighbors is an example- or instance-based approach to interpretability. Since it uses no modular learning of any global structure to make predictions there are also no global weights or decisions to interpret in the first place. The algorithm, in the case of classification, classifies the instance depending on the most common class in the neighbourhood of the point in the feature space. The neighbourhood then depends on the number of neighbours and the distance metric we use to identify the nearest k neighbours. To explain a specific prediction, we rely on the nearest neighbours used and their common feature patterns. Published work has used learned local feature relevance matrices to identify which dimensions matter most for each prediction's neighbourhood [73]. This implies, however, that in cases of higher dimensionality, seeing those patterns becomes increasingly difficult and the explanations less readable.

Making Deep Learning Inherently Interpretable. Static or tabular data are not the only type of data encountered in clinical machine learning applications. Often, images or timeseries require a different *modus operandi*, and deep learning, a machine learning paradigm relying on neural networks with a large number of layers (convolutional, recurrent, or otherwise), has proved very effective at learning from this data. Traditional machine learning models require feature engineering, especially when the inputs are images, whereas deep learning, through representation learning, can learn from raw data directly the optimal representation of internal features [115]. The challenge is making more complex and predictive models interpretable without sacrificing their performance.

Fuzzy logic. Causality and fuzzy logic can help make combined systems with deep learning inherently interpretable. Traditional machine learning methods have seen their own implementations in the fuzzy logic realm but the focus here is on transforming inherently uninterpretable deep learning by fuzzy rules. For decision tree, support vector machine, and nearest neighbour fuzzy implementations, consult the following literature: [24, 94, 119, 122, 132]. The assumption in fuzzy logic is that instead of assuming input or output is deterministic, i.e. fixed value, one assigns uncertainty to both ends of the learning system. An example would be the decisions that a human makes while driving a car as provided by [40]. The decision-making of "if the distance to the car ahead is less than 2.5 m and the road is 10% slippery then reduce car speed by 25%" is approximated with numerical uncertainty reflecting the imprecise language of *If* the distance to the car ahead is low and the road is slightly slippery *Then* slow down. The numerical meanings of "low", "close", and "slow down" can vary and are, thus, fuzzy.

Breaking down the inference problem into fuzzy logic if-then steps helps interpretability and can be combined with high-performing opaque deep learning models through, for example, a fusion layer at the end of a neural network, one part of which is deep representational learning and the other having fuzzy rules [51]. The weights in the fuzzy part of the networks are themselves determined by fuzzy parameters, which can aid in modelling uncertainty and improving prediction performance [155]. Another approach is to add the fuzzy logic system as a module with autoencoder structures through pre-trained layer replacement, often referred to as a deep type-2 fuzzy logic system (D2FLS) [41]. Regardless of strategy, fuzzy if-then rules can open up black-box deep learning models while keeping the superior predictive performance. There are limitations, however, as to how well suited these methods are for interpretability itself. Since they were built for uncertainty integration, they have not been fully optimised for explainability applications, especially in higher dimensions, though hybrid approaches combining deep learning with type-2 fuzzy logic systems have begun to address this challenge [39]. Future work should test these methods across diverse tasks, from static tabular data to image classification, and tackle higher dimensionality through feature grouping.

Boolean rules. Using logic-based rules to interpret complex models without external add-ons also includes work on Boolean rules generation with column generation methods (BRCGs). These methods reduce binary classification problems to a set of compact Boolean clauses similar to the ones seen earlier in rule-based models and fuzzy logic but with a set structure of more compact and easier-to-explain rules [47, 50, 114]. Each conjunction in these unordered rule sets is considered an individual rule, and a positive prediction occurs when at least one of the rules is satisfied. These rules can be mined from the data (with decision trees, for example) after which they can be selected through methods such as integer programming though this often comes with a limited search space [50]. Rule selection through column generation searches over an exponential number of all possible clauses optimised in a greedy manner with the combined reduced cost objective. This allows for a quantitative way of measuring interpretability by just looking at the size of the rule set itself. In cases of higher dimensionality, this approach becomes less tractable and an approximate Pricing Problem is used by randomly selecting a limited number of features and samples [50]. The classification accuracy and computational costs for these binary classifiers are, however, a large limitation to their widespread acceptance. Recent work has attempted to use optimised column generation to aid with the computational constraints and has included fairness in the objective evaluation through equality of opportunity and equalised odds constraints [114].

Transfer learning. Besides relying on fusion layers and rule-based methods to turn opaque models inherently interpretable by some type of combination, another approach is using transfer learning for explainability. Transfer learning uses pre-training or learning weights for one task and 'transfers' that knowledge and the learned weights to another problem, sometimes with a limited amount of re-learning and often to a different domain [224]. By using transfer learning to preserve the high predictive performance of black-box models and transferring them to interpretable, simpler models, we could balance between deep learning and transparency. The limitation, however, is on how transferable the features learned from the complex model are to the simpler model [219]. Assuming sufficiently transferable features, through this approach, we could simultaneously use a simpler and more reliable model that clinicians would be familiar with while preserving higher performance, apply a model trained on a large amount of data to a smaller dataset thereby avoiding overfitting, and reduce computational costs by simply 'cut-and-pasting' a pre-trained model [54, 224]. All of these can help achieve more explainable systems that are not just interpretable, but easier to check for fairness and bias, especially considering the smaller and more accessible datasets that can now be used for deployment.

Methods like ProfWeight transfer the learning process from a pre-trained deep neural network with a high predictive capability to a simpler interpretable model or a very shallow network of low complexity and low test accuracy. The method relies on flattened intermediate representations used to generate confidence scores of linear probes for weighting of samples during training of the simpler models. This approach has not yet been sufficiently evaluated in real-world applications such as clinical risk prediction but could hold great promise in achieving a stable balance between higher test accuracies and white-box models [54]. Possible extensions could include new and more challenging data domains and applications, experimenting with a combination of different models of varying complexity, and optimising for inference time and reducing costs since the initial weights still need to be computed from a complex deep learning model.

3.3 Ante-hoc vs. Post-hoc

We have previously described several methods underpinning inherently interpretable models in AI, including decision trees and rule-based models. Often, because these models do not require any post-prediction or post-inference interpretability, they get classified as ante-hoc methods. In cases of models that are not IIM, such as deep learning models or large ensembles, a post-hoc explanation framework needs to be applied after post-model learning. The post-hoc category splits into two subgroups, those post-hoc methods that only work for specific data modalities or models, and model-agnostic explainability methods. The post-hoc approach addresses the trade-off between high interpretability and high predictive performance by providing add-on interpretability to complex but high-performing models. Thus, instead of developing an IIM which, as we have seen, suffers under certain constraints either as part of the assumptions or model prediction capabilities, post-hoc explanations can be paired with a wide swath of complex AI models to provide interpretability without understanding the inherent reasoning of the model itself.

Making Neural Networks Ante-hoc Interpretable. Recent work on making deep neural models ante-hoc interpretable includes approaches like self-explainable neural networks (SENNs) and CoFrNets. SENNs rely on progressive generalisation of linear models as a function of locally interpretable functions and representations of the input. An illustration of the model architecture is provided in [10]. SENNs have been tested on a few tabular and low-complexity image datasets. CoFrNets, on the other hand, rely on the mathematics of continued fractions

(CFs), typically represented as a ladder-like sequence: $a_0 + \frac{b_1}{a_1+\frac{b_2}{a_2+\cdots}}$ which can represent any real number and any analytic function [44, 99]. Using the repeating structure and linear functions between layers (which can be shown to be sufficient to achieve universal approximation), the input is passed to each layer and a linear number of weights is learned for each layer instead of the quadratic amount in classic neural networks [165]. Per-example feature importance is then computed using continuant-based closed-form expressions as a function of the input and global attributions taking advantage of the ladder being a representation of a multivariate power series with the coefficients of the two forms mapping one-to-one. While competitive with interpretable models and MLPs, further work is needed to close the gap with state-of-the-art black-box models.

Post-hoc Methods. Post-hoc methods can provide interpretability and limited explainability after model training, inference, and deployment. These methods can be categorised into attribution-based and non-attribution approaches, each with advantages for clinical applications.

Shapley values. These include feature attribution (global and local) methods like Shapley values which adapt concepts from game theory to investigate feature importance and contributions to predictions. Shapley values can be seen as a fair way to distribute credit among team members, in this case, features. A Shapley value is calculated for each feature j with feature value x_j by considering all possible subsets of the m features excluding feature j. The value is the difference between the results of the characteristic function v on M (the set of all features) and S (the subset of M without feature j). The Shapley value is then averaged across the marginal contributions of all possible combinations of the feature unions:

$$\varphi_j(v) = \sum_{S \subset M \setminus \{j\}} \frac{|S|!(m-|S|-1)!}{m!} \left(v\left(S \cup \{j\}\right) - v(S)\right)$$

Shapley values are useful for both finding the contribution of each feature to an individual prediction as well as providing global explainability across all samples [88, 125]. Calculating the value itself is computationally expensive even on moderately sized datasets. KernelSHAP addresses this by defining a specific kernel that reduces the number of required coalition samples through a combination of sampling and penalised linear regression. KernelSHAP, however, does not account for feature dependence, as it marginalises over absent features independently, which can lead to model evaluation on unrealistic, out-of-distribution inputs. TreeSHAP partially addresses this by using conditional expectations, but can assign non-zero importance to features with no true predictive influence when those features are correlated with important ones [140].

A step-by-step SHAP implementation for EHR risk prediction, covering choice of variant (TreeSHAP, DeepSHAP, KernelSHAP), local and global plots, and a clinical validation checklist, is provided in Listing S2 of the supplementary material.

Fairness-Aware Practice: When implementing SHAP for clinical risk models: 1. Stratify explanations by race, ethnicity, sex, and socioeconomic indicators 2. Flag features that show >20% importance disparity across groups 3. Validate that high-importance clinical features are equally measurable across populations (e.g., some biomarkers have different reference ranges by ancestry) 4. Document which protected attributes were considered and why

LIME. Local interpretable model-agnostic explanations (LIME) focuses only on explaining local or individual predictions. LIME relies on generating perturbations around the instance being explained and observing how they affect the corresponding predictions given an interpretable model like some of the IIMs mentioned earlier while approximating the black-box model in the neighbourhood of the sample [170]. In clinical risk prediction, LIME often produces unstable feature rankings because of its random perturbations [14, 220]. [220] propose DLIME (D standing for deterministic), which uses hierarchical clustering to group the training data together and kNN to select the relevant cluster of the new instance being explained instead of relying on randomness. Evaluated on clinical risk problems, DLIME is a more stable version of instance-wise interpretability.

Practical Tip: When using LIME for EHR data, consider DLIME for more stable explanations, especially when dealing with temporal patterns in patient histories.

Local Interpretable Visual Explanations (LIVE) build on the LIME paradigm by prioritising the visualisation of the interpretability results while keeping the random data perturbations approach and local point variation. While LIVE does not require an interpretable input space since it approximates the black box model directly from the data, it has no theoretical foundation like Shapley values. BreakDown uses a greedy approach to post-hoc, model-agnostic interpretability that sequentially attributes portions of an individual prediction to input features [186]. Starting from the instance prediction $f(\mathbf{x}_{\text{new}})$, variables are conditioned one at a time to trace a path to the average expected prediction $\mathbb{E}[f(\mathbf{x})]$, with each step's contribution attributed to the corresponding feature. These methods, however, decompose final prediction into additive attribution components and usually do not work well for models with significant inter-feature interactions.

A worked LIME implementation, including a stability check across random seeds and a clinical validation protocol, is provided in Listing S3 of the supplementary material.

Contrastive explanations. Manipulating feature input and then measuring its effects on the output is a common set-up of many interpretability methods. The contrastive explanations method (CEM) finds what features should be minimally and sufficiently present for classification and analogously what should be minimally and necessarily absent. This method was evaluated on diverse datasets including brain activity data by generating explanations of the form "an input x is classified in class y because features $f_j, \cdots, f_k$ are present and because features $f_m, \cdots, f_p$ are absent" [53]. As compared to other methods like LIME, CEM highlights what features need to be present for a specific classification and not just list positively or negatively relevant features that may not be necessary or sufficient. Finding this set of features is an optimisation problem where an autoencoder can optionally be used as a regularisation term to ensure that perturbed examples remain close to the data manifold. These types of explanations might be more intuitive to many clinical applications especially in diagnosis as highlighted by [138, 139]. Finding contrastive explanations can be done with an adversarial model, for example, but the original CEM was limited to simple datasets such as greyscale images. [130] leverage latent features from generative models to produce contrastive explanations on richer data including medical images. However, the extension of contrastive explanations to complex clinical data modalities such as multivariate time series from electronic health records remains an open challenge, and the idea of "minimal absence" is less clearly defined for temporal sequences than for spatial features in images.

Prototype-based methods. Prototype-based methods explain model behaviour by finding representative examples from the data like the clinical reasoning pattern of comparing a new case to previously observed ones. ProtoDash is a method that generates prototypes from the data and assigns non-negative weights to each of them depending on their importance to the prediction problem. Since each set of prototypes can be weighted, ProtoDash can better represent skewed data distributions, a property that is particularly relevant in clinical settings where class imbalance is common [145]. ProtoDash is an instance-based interpretability method as the prototypes it selects are representative examples from the data distribution rather than feature-level attributions. Other earlier instance-level methods like K-medoids and MMD-critic [104] tried to do the same, but ProtoDash produces more general example-based explanations with higher prototype diversity that can help reveal the reasons for relatively low test accuracy and maintain consistent interpretability [152].

3.4 Gradient-Based and Visual Attribution Methods

So far we have only discussed methods that manipulate the data space in some form to evaluate feature or example importance for prediction, but other post-hoc paradigms exist that more directly measure model learning.

In deep learning, learned gradients are used to update weights through the learning process. By computing the gradient of the model output with respect to the input, we can construct a saliency map that highlights which parts of an individual example, such as a region of an image or a segment of a timeseries, are most influential for the model's prediction [182]. Examples include using visual saliency maps to highlight sections of ECG signal important for arrhythmia classification [100] or chest radiographs for detecting pathologies [16]. These saliency maps are often sensitive to input noise, and researchers have proposed several remedies, including adding noise to the data multiple times and then taking the average of the resulting saliency maps for each example, as with SmoothGrad [183]. Newer methods like saliency-guided training iteratively mask features with low gradient values and then minimise a combined loss function of the KL divergence between model outputs from the original and masked inputs, and the model's own loss function. Experiments have shown the latter maintains higher predictive performance and reduces noise sensitivity [89].

General limitations of saliency maps in clinical risk prediction include their tendency to highlight information that is not clinically meaningful, as demonstrated by sanity checks showing that some methods produce similar outputs regardless of model parameters or training data [96, 174], and evaluations in medical imaging where saliency-based localisation has underperformed dedicated localisation approaches [16]. Localising the relevant area of an input does not explain why that area is important for the model's prediction, which can appear misleading to domain experts. Saliency methods have also been shown to be vulnerable to adversarial manipulation, where small input perturbations can dramatically alter the generated explanation while leaving the model's prediction largely unchanged [68].

Attention Mechanisms. The underlying heuristic of attention mechanisms is to visualise weights from the learning process to highlight which aspects of the data the model focuses on for training. The wide popularity of this mechanism is due to it avoiding the typical trade-off between predictive performance and interpretability, and achieving strong results on both fronts. An attention function maps a query and a set of key-value pairs where the output is computed as a weighted sum of the values and the weights are computed by a compatibility function of the query with the corresponding key. Multi-head attention parallelises the attention layers to project multiple mappings simultaneously. A comparison of the two frameworks can be seen in [197]. Compared to recurrent layers requiring $O(n)$ sequential operations to process a sequence, a self-attention layer connects all positions with a constant number of sequentially executed operations [197]. However, the per-layer computational complexity of self-attention is $O(n^2 \cdot d)$ versus $O(n \cdot d^2)$ for recurrent layers, so self-attention is faster when the sequence length n is smaller than the representation dimensionality d.

Implementing Attention for Time-Series Clinical Data

In the example of timeseries data, the first step is to initialise the attention vectors mapped to the features such that for each feature j an attention vector $\boldsymbol{a}_j$ of length T (number of time-steps) is learned with $|\boldsymbol{a}_j| = 1$:

$$\boldsymbol{X}_{\text{new}} = \boldsymbol{A} \odot \boldsymbol{X}$$

where $\boldsymbol{X}$ represents the $T \times m$ input data (m number of variables). Once softmax normalisation is applied, $\boldsymbol{a}_{jt}$ can be interpreted as a contribution of feature j within a fixed time step t. [66] extend this for global interpretability by applying the softmax to the transposed input with the same notation:

$$\boldsymbol{a}_t = \text{softmax}\left(\boldsymbol{x}_t \boldsymbol{W}_t\right)$$

By aggregating values $\boldsymbol{a}_{j1}, \ldots, \boldsymbol{a}_{jT}$ through time one can get the global contribution of the j-th feature $\boldsymbol{x}_t$ and weights $\boldsymbol{W}_t$ at time t. Not only does the method provide an absolute ranking of features, we can use the softmax activation matrix to extract patient-level feature importance as well as visualise the attention weights using heatmaps (attention maps) both in time-series and imaging problems, similar to the saliency maps scenario despite criticism of the lack of explainability of such maps for standard attention mechanisms [93].

A PyTorch reference implementation of a clinical temporal attention module, with extraction, visualisation, and validation steps, is provided in Listing S4 of the supplementary material.

Since being originally proposed as an alternative to convolutional and recurrent behaviour in deep learning pattern recognition, transformers (models based solely on attention mechanisms) have become robust prediction models in various applications for clinical risk prediction, including ICU outcome prediction, ECG signal diagnosis, and blood pressure response [85, 101, 117, 197, 213]. Some limitations include inability to distinguish between positive and negative associations among features and output, lack of propagated relevancy through the attention layers partially addressed through a new layer-propagation strategy, and unstable interpretability across varying data conditions [21, 37].

This section has walked through a range of approaches to interpretable deep learning, from purpose-built ante-hoc glass-box architectures to widely adopted post-hoc tools like LIME and SHAP. The takeaway is that no single method is a silver bullet as most remain sensitive to random noise, permutation, and covariate shift, and very few have undergone rigorous clinical evaluation of their interpretability claims. Before deploying any of these techniques in a clinical pipeline, we recommend pairing at least two complementary methods (e.g. attention maps with perturbation-based validation) and evaluating the resulting explanations against domain expertise, as outlined in the validation steps throughout this tutorial.

3.5 Global vs. Local

Explainability methods also divide into global (feature-space) and local (sample-space) levels. A local or instance-based (used synonymously for some methods) explanation explains the prediction of an individual sample, whereas a global explanation attempts to provide a summarised view of the entire sample set. In clinical risk prediction, both can be valuable. Local explanations can be used for explaining individual patient trajectories and predictions, for example, while global explanations can provide a more holistic view of the impacts of certain risk factors on a population-size patient cohort [49, 60].

- Use **local explanations** when: explaining individual patient diagnoses, discussing treatment options with patients, or investigating edge cases
- Use **global explanations** when: understanding population-level risk factors, identifying subgroup-level differences, validating model behaviour against clinical knowledge, or regulatory compliance

Several of the methods discussed in earlier sections support one or both of these levels, and recognising which mode a given method operates in is important for matching explanations to their intended audience. From the interpretability methods covered earlier, LIME and SHAP also support local explanations by providing sample-level explanations. In the case of LIME, we generate perturbed examples in the neighbourhood of the sample and fit a local surrogate model while SHAP guarantees a fair allocation of feature contributions satisfying local accuracy, missingness, and consistency properties. SHAP provides a principled attribution even when features are correlated, though at potentially higher computational cost while providing a theoretical framework to prove, under certain assumptions, the unique existence of such local models [125, 194]. An extension of LIME which resolves the significant dependence on weights assigned to different perturbed samples is ILIME which uses the influence of perturbed instances on the instance of interest as well as their Euclidean distance to estimate the weights with influence functions [59].

ProtoDash produces as an explanation representative examples or prototypes from the data that estimate the underlying distribution. As mentioned earlier, ProtoDash generates these samples by assigning non-negative weights of importance. The generation depends on the property of weak sub-modularity of its scoring function to find samples efficiently. The computational cost of such guarantees is significant, thus a class of approximate weak sub-modular functions are used [79]. Prototype-based explanations are typically considered a form of local or example-based explanation, though they can serve a global summarisation purpose when the full set of prototypes is considered together.

Other methods mentioned earlier like ProfWeight, Boolean rules, and others all provide global explainability exclusively which means they provide feature importance and explanations across the entire sample set without resolution to the sample level. We can then say that such methods provide an aggregate estimate explanation for the model and the data and are useful in understanding general behaviours of the system.

3.6 Model-Agnostic vs. Model-Specific

Explainability methods can also be model-agnostic, meaning that they can be used to provide interpretable results for a variety of models. This distinction matters in complex settings, where we might deploy multiple models across different tasks, a gradient-boosted model for tabular EHR-based risk prediction alongside a convolutional network for radiological imaging, for example. Model-agnostic methods allow a consistent explainability framework to be applied across these heterogeneous tools, which simplifies transparency requirements in regulated domains like healthcare. This is the case for SHAP, for example, which provides feature-level attributions regardless of the underlying model, from extreme gradient-boosted decision trees to neural networks [125]. Model-specific variants such as TreeSHAP further exploit known model structure for computational efficiency, which can matter in time-sensitive clinical environments such as ICU decision support [126, 129]. Permutation importance is another widely used model-agnostic approach that measures each feature's contribution by randomly shuffling its values and observing the resulting drop in model performance [141]. Because it treats the model as a black box, it can be applied to any supervised learning algorithm. However, two limitations might be relevant in clinical contexts. First, it can produce misleading results when features are correlated, a not uncommon occurrence in EHR data. Shuffling one correlated feature may have little effect on performance because the model can still rely on its correlate, leading to underestimation of both features' importance [141]. Second, while its computational cost scales linearly with the number of features rather than combinatorially, it still requires repeated model evaluations for stable estimates, which can become burdensome for large clinical datasets with many input features. In practice, the choice between model-agnostic and model-specific methods might depend on the deployment context where model-agnostic approaches are preferable when comparing candidate models during clinical validation, while model-specific methods may be favoured once a particular architecture has been selected and computational efficiency or explanation fidelity becomes the priority.

Visualisation Methods for Model-Agnostic Explanations. Some model-agnostic methods are focussed on visualisations, such as partial dependence plots (PDPs) which rely on using a partial dependence function of the output with respect to subsets of the feature space. Values in the input space are varied along a grid of values spanning the feature's domain, where for each grid value, the model's predictions are averaged over the observed values of all other features with respect to their marginal distribution. The partial dependence function is by approximated through Monte Carlo averaging of the trained ML model's predictions over the training data from which plots of associations between covariates and output are created [65]. The assumption is that because this is done for a single predictor at a time, that predictor is strictly uncorrelated. Due to substantial interaction effects, the partial response relationship can be heterogeneous, and an average curve such as the PDP can obscure the complexity of the modelled relationship, which led to individual conditional expectation (ICE) plots. In ICE, instead of plotting the feature average partial effect on the output, one plots the estimated conditional expectation curves, each reflecting the output as a function of the feature effectively disaggregating divergent effects. The PDP curve in the plots is then simply the average of N ICE curves behaving as a local interpretability model that plots a curve for each instance of the data [72]. Some limitations of ICE plots include their overcrowding due to curves created for each sample instance in the plot, difficulty detecting outliers and overfitting extrapolations visually. Furthermore, the complexity of a higher feature space makes this method practically infeasible. Some recent advances like ICE feature impacts address these by allowing comparisons between different ICE plots and defining a metric, taking into account all samples as well as outliers over which feature contributions are averaged to identify the most impactful features (not the same as PDP where the curves and expectations are themselves averaged for each covariate contribution), thus severely decluttering the plots [215].

3.7 Attribution-Based Methods

Attribution methods give importance scores to input features for a given prediction. They apply across modalities, tabular EHRs, medical images, and clinical time series, though specific techniques suit some data types better than others.

Gradient/backprop (DL-specific). Integrated Gradients accumulates gradients along a path from a chosen baseline to the input, satisfying both the sensitivity and implementation invariance axioms [189]. DeepLIFT does something different as it propagates activation differences layer-wise, which gives it greater stability than raw gradients when units are saturated [181]. Layer-wise Relevance Propagation (LRP) redistributes the output score back through the network under conservation constraints and produces class-consistent heatmaps [143]. SmoothGrad takes a simpler route, averaging gradients over multiple noisy copies of the input to cut visual noise in saliency maps [183]. DeconvNet and Guided Backpropagation also produce visualisations using modified backpropagation rules [185, 221], but randomisation tests have shown that their outputs can decouple from what the model actually learned [4].

Activation mapping (vision DL-specific). CAM and Grad-CAM localise class-discriminative regions by weighting final convolutional-layer activations, with Grad-CAM++ extending the idea to handle multiple object instances better [36, 177]. In radiology, these heatmaps can serve as a quick check on whether a model is looking at plausible anatomy or latching onto spurious image features [191].

Attribution methods are helpful as post-hoc, but they have known pitfalls. Saliency maps can fail sanity checks, remaining visually unchanged even after model weights are randomised [4]. With tabular EHRs, correlated features like creatinine and eGFR can distort additive attribution scores. And for Integrated Gradients specifically, a poorly chosen baseline (all-zeros, say, in a domain where zero is not meaningfully "absent") will produce misleading importance maps.

Practical Tip: For tree models on EHRs, prefer TreeSHAP for speed and consistency; group highly correlated labs (e.g., creatinine/eGFR) or report SHAP for feature *groups*. For deep imaging/time-series, (i) run model/label randomisation sanity checks; (ii) compare two complementary methods (e.g., Grad-CAM and RISE); and (iii) have a clinician rate alignment with known risk factors (e.g., cardiomegaly regions for HF risk) before trusting explanations in workflows.

A PyTorch implementation of Grad-CAM for medical imaging, with a radiologist-facing validation protocol, is provided in Listing S5 of the supplementary material.

3.8 Perturbation vs. Gradient vs. Instance-Based Approaches

Some classifications of explainability methods are based on how attributions are computed relative to the input-output relationship. The most common grouping is between gradient-based and perturbation-based approaches.

Perturbation-based methods. Perturbation-based methods modify the input to a fixed model and observe how the output changes, thereby inferring the importance of individual features for a given prediction [9, 11, 90]. In the image domain, perturbations are typically performed by occluding or replacing pixel or patch values to produce saliency maps at varying levels of granularity. Pixel-wise perturbations are more spatially discrete and localise salient regions more precisely, but their finer granularity leads to poorer representation of the semantic boundaries of salient objects [90]. Patch-wise methods smooth over fine spatial detail but give saliency maps whose boundaries correspond more naturally to object contours. In both cases, selecting an appropriate scope and shape of perturbation across diverse inputs remains an open challenge.

Gradient-based methods. An example of the gradient-based approach is integrated gradients that can help visualise input importance by calculating the integral of the gradients of the output prediction with respect to the input image pixels without any modification to the original network [189]. Integrated gradients are defined as the path integral of the gradients along the straight line path from the baseline $\boldsymbol{x}'$ to the input $\boldsymbol{x}$. The integrated gradient along the j^{th} dimension (can be understood as j^{th} feature) for an input $\boldsymbol{x}$ and baseline $\boldsymbol{x}'$ with the interpolation parameter that traces the straight-line path from baseline to input of α defined as follows:

$$\mathrm{IG}_j(\boldsymbol{x}) ::= \left(\boldsymbol{x}_j - \boldsymbol{x}'_j\right) \times \int_{\alpha=0}^{1} \frac{\partial F\left(\boldsymbol{x}' + \alpha\left(\boldsymbol{x} - \boldsymbol{x}'\right)\right)}{\partial \boldsymbol{x}_j} d\alpha \tag{3}$$

Another sub-domain of gradient-based methods is saliency-based methods which use gradient values either raw or normalised to infer salient features like in saliency maps described earlier. [11] highlight the weakness of such methods under robustness checks despite their advantages like simple formulations and lack of reliance on model access. Saliency or heatmaps have also been criticised for insufficient correlation with the network which they are meant to interpret thereby undermining their reliability as interpreters of the model. To address this concern, [166] propose Integrated-Gradients Optimized Saliency (I-GOS), a method which optimises a heatmap with the objective of maximally decreasing classification scores on the masked image. This is done by computing descent directions based on integrated gradients thus avoiding local optima and speeding up convergence. Gradient-based methods are often, however, not invariant under simple transformations of the input, and are very sensitive to the choice of reference point.

Other related attribution techniques include deconvolution approaches and Deep Learning Important FeaTures (DeepLIFT) [181]. Deconvolutional networks (Deconvnet), introduced by [221], reconstruct input-space patterns by mapping neuron activations in a given layer back to the input, highlighting the regions that most strongly activate a given neuron [185]. Because deconvolution-style methods zero out negative gradients, they tend not to capture inputs that negatively contribute to the output prediction [181]. DeepLIFT was specifically designed to mitigate this limitation by propagating differences in activations relative to a reference, rather than relying on raw gradients alone.

Layer-wise relevance propagation. Layer-wise Relevance Propagation (LRP) is an attribution method that propagates the prediction backwards in the neural network and compared to its perturbation- and gradient-based competitors does not involve multiple computationally expensive neural network evaluations. It instead uses the graph structure of the deep neural network to generate explanations [143]. The relevance score that is propagated is estimated for neurons j and k, for example, based on those above them:

$$R_j = \sum_k \frac{z_{jk}}{\sum_j z_{jk}} R_k \tag{4}$$

where $z_{jk} = a_j w_{jk}$ represents the weighted activation, capturing the extent to which neuron j contributes to neuron k (the denominator ensures that relevance is conserved as it propagates downward). At the output layer, the relevance score is initialised as the logit corresponding to the target class, obtained from a single forward pass. A key property of LRP is that the total relevance is conserved across layers, remaining constant throughout propagation. LRP has been applied in many settings to offer interpretability to deep learning models which will be highlighted later under the applications section. LRP has been shown to give better relevance or attributions than those by DeepLIFT without relying on the need for a reference input [13, 75]. LRP suffers from its dependence on ReLU activations resulting in non-negative attribution maps thereby limiting the interpretability of the application, later addressed by an extension of the transformer using LRP-type of relevance score to compute the relative relevance of heads in the attention mechanism using GELU instead. Class-specific interpretability can be obtained by extensions like Contrastive-LRP (CLRP) and Softmax-Gradient LRP (SGLRP) where the results of the selected class are compared with the results of all other classes, to emphasise the differences and produce a class-dependent heatmap [37, 91].

Other methods rely on manipulating the layers of the neural network itself similar to attention layers. Class Activation Mapping (CAM) removes the last fully connected layers and replaces the MaxPooling layer with a global average pooling (GAP) layer after which the weighted average of features is extracted to form a per-class activation map (overlaid on the input like a heatmap) [223]. This makes it a non-gradient-based method but extensions of it such as Gradient-weighted CAM (Grad-CAM) and Grad-CAM++ which propagate gradients through the network in obtaining the final heatmaps for input regional relevance could be classified as gradient-based. Other examples of the method include methods like Eigen-CAM which extracts the principal components from the convolutional layers and helps against classification errors made by fully connected layers in CNNs without needing to backpropagate gradients or use feature weighting [146, 203].

Both perturbation- and gradient-based methods can be described as attribution-based, in that they establish a mapping from regions or features of the input to the output prediction. A practical drawback of perturbation-based methods is the combinatorial explosion of possible input modifications if one attempts an exhaustive search without approximation. Additionally, different samples of the same class may yield contradictory explanations. Gradient-based methods avoid this by using the gradient as a local approximation of how the output changes with respect to the input, avoiding the need to evaluate numerous feature subsets. While this makes them faster, some gradient-based saliency methods have been shown to fail sanity checks, producing explanations insensitive to model parameters [4, 13]. An advantage of perturbation-based methods is their model-agnostic nature, they only require the ability to query the model repeatedly with modified inputs [90].

Instance-based methods. Instance-based methods attempt to remedy the disadvantages of perturbation- and gradient-based methods by optimising for the fidelity of instance-level explanations. An example are amortised explanation methods (AEMs) which generate explanations by learning a global selector model that extracts locally important features in an instance of data with a single forward pass using an objective that measures the fidelity of the explanations. AEMs assess these selections with a predictor model for the output. L2X and INVASE fit the predictor and selector models jointly, often referred to as joint amortised explanation methods (JAMs) [95]. Since they are instance-based, JAMs have been previously deployed to explain mortality predictions at a patient-level [218]. Some problems with JAMs include their propensity to encode predictions with the learned selector and failure to select features involved in control flow (features involved only in branching decisions/nodes of tree structured generative process).

An example of this could be mortality predictions for patients with chest pain using EHRs as [95] point out. Blood troponin levels can be a control flow feature where abnormal values indicate that cardiac imaging should be used to assess severity whereas normal values indicate that the chest pain may be non-cardiac. In this case, using JAMs to interpret the prediction will not represent the true role of troponin in patient mortality.

To address these issues, [95] propose REAL-X and EVAL-X, two frameworks that combine efficient instance-level explanations with a single forward pass while also detecting when predictions are encoded in explanations without making out-of-distribution queries using approximations of the true data generating distribution. Other examples of instance-based methods include ProtoDash, influential instances, MMD-Critic, and counterfactuals described in detail in earlier sections.

3.9 Prototype-Based Approaches

Prototype methods explain a model's prediction by comparing inputs to learned representative examples, prototypes. Nauta et al. [151] integrate prototypes into a decision tree where each internal node contains a trainable prototypical part, and the model follows a clear path of "looks like that" comparisons. This helps them create a globally interpretable architecture with compact explanations that match non-hierarchical prototype methods in accuracy. When prototype networks are paired with Vision Transformer (ViT) backbones, they can often suffer from "distraction", where prototypes are disproportionately activated by background regions. Xue et al. [214] address this with ProtoPFormer, which introduces both global and local prototypes. The global prototypes guide the local ones toward foreground regions and improve interpretability. Donnelly et al. [56] propose Deformable

ProtoPNet, which augments prototypes with spatial flexibility so that their constituent parts can shift relative positions depending on the input. This could help with more coherent prototype matches in clinical imaging, where anatomical structures vary across patients.

Prototype learning can also be improved by rethinking how prototypes relate to the classification boundary. Wang et al. [206] draw an analogy between prototype learning and support vector machines, identifying that standard ProtoPNet prototypes lie far from the classification boundary (trivial prototypes). They propose additionally learning support prototypes near the boundary, and combine both types in ST-ProtoPNet to improve classification accuracy and localisation. Earlier, Rymarczyk et al. [172] introduced prototype sharing across classes via data-dependent merge-pruning, substantially reducing the number of prototypes while preserving accuracy. For text, Hong et al. [83] introduced ProtoryNet, which finds the most similar prototype for each sentence in a document and tracks the resulting prototype trajectory over time, providing sentence-level, case-based explanations.

3.10 Concept Learning

Concept learning aims to explain model behaviour in terms of high-level, human-understandable factors ("concepts") rather than raw features or pixels. Instead of attributing importance to individual inputs, the model is probed or trained to use a small set of concepts that clinicians can reason about (e.g., "cardiomegaly", "ST-elevation"). Compared with attribution maps, concept methods support *why*-style arguments and clinician intervention. Concepts can be (i) *post-hoc* and external to the model, learned from examples and then probed in hidden layers; or (ii) *ante-hoc*, embedded in the model with architectural constraints so that predictions depend explicitly on concept activations. One recent modular design attaches a concept encoder to an existing backbone and adds an optional reconstructor that measures how faithfully the learned concepts capture the input; auxiliary components can be dropped at inference [175] (Figure 2). Core models can be classified into:

(1) TCAV (Testing with Concept Activation Vectors): learns linear directions for user-provided concepts and measures a model's directional sensitivity to each concept via gradients [105]. Requires curated, labeled concept sets.
(2) ACE: automatically mines candidate concepts (superpixels/patch clusters) and evaluates their importance, reducing manual curation bias [69].
(3) ConceptSHAP: assigns each concept a Shapley-value score reflecting its contribution to the completeness of the concept set, the degree to which the concepts collectively recover the model's predictions, thereby providing additive, axiomatic attribution [216].
(4) Concept Bottleneck Models (CBMs): a model first predicts a set of concepts and then predicts the label from those concepts; clinicians can inspect and override concept predictions at test time [108].

Recent surveys position concept methods as a distinct mechanism in clinical XAI, complementary to attributions and counterfactuals [84].

To move beyond correlation, CaCE estimates the causal effect of adding/removing a concept on the prediction by generating counterfactuals (often with conditional VAEs) and averaging output deltas [74]. CaCE helps disambiguate correlated concepts (e.g., "pleural effusion" vs. "atelectasis"), but depends on the quality and assumptions of the generative model. Unsupervised or weakly supervised pipelines can provide candidate concepts through dimensionality reduction, clustering, or patch mining, then validate them against clinical reasoning.

Concept learning fits settings where stakeholders need semantically meaningful factors, the ability to edit or constrain model reasoning (as in CBMs), or population-level auditing for spurious features. Attribution methods remain preferable when fine-grained spatial localisation is most important and no natural concept vocabulary is available.

Concept leakage (concepts inadvertently encoding the label), collinearity among overlapping concepts, incomplete concept vocabularies, and generator bias (in CaCE's counterfactual synthesis) all limit concept-based methods in practice. Mitigations include concept set audits with domain experts, reporting concept completeness [216], sensitivity to replacements (swap concepts with near neighbours), and counterfactual validity checks [74].

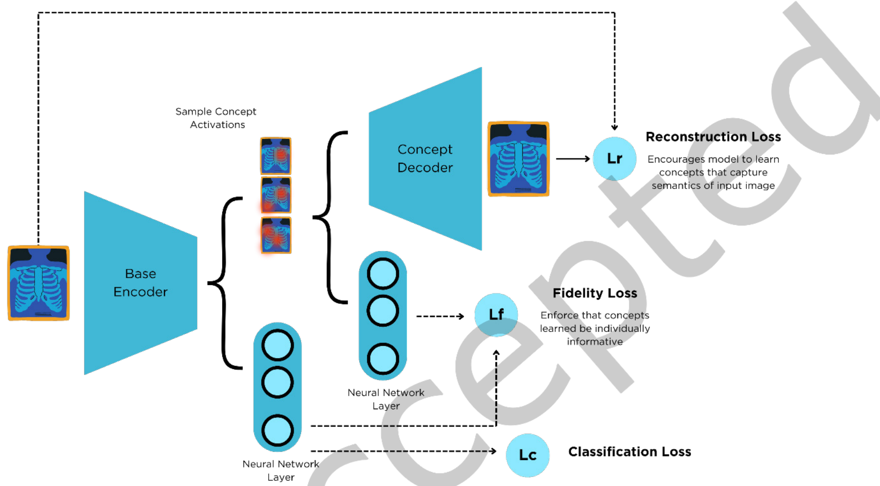

Fig. 2. A modular concept-learning framework that appends an interpretable concept encoder (and optional reconstructor) to a backbone; most auxiliary components can be discarded at inference [175].

Concept-level reasoning has also been adapted to text. INLP removes linearly decodable attributes from hidden representations, originally to audit bias but applicable more broadly to probing for learned concepts [168]. CausaLM extends this idea to causal concept editing through counterfactual language modelling [62]. On the evaluation side, the CEBaB benchmark frames concept effects as treatment effects and applies general-purpose meta-learners such as the S-learner [112] as baselines, revealing gaps where approximate counterfactual methods can outperform dedicated explanation techniques at recovering human-judged causal directions [2].

Practical Tip: In clinical workflows, we can pre-register a small concept set (10–30) with clinicians, then: (1) measure *concept completeness* (ConceptSHAP) to ensure coverage [216]; (2) run *negative controls* (near-miss concepts) to detect leakage; (3) perform *CaCE spot-checks* on a stratified validation cohort to confirm causal directions [74]; and (4) if intervention is required, deploy a CBM head where clinicians can correct concept predictions at the point of care [108].

3.11 Counterfactual Explanations

Counterfactual explanations mentioned earlier are a more general and abstract way to explain models while addressing fairness and bias concerns also can be classified as a local interpretability method. In short, "a counterfactual explanation of a prediction describes the smallest change to the feature values that changes the prediction to a predefined output" as explained by [140].

Creating these explanations for machine learning models was first proposed by [205] who suggest minimising the loss:

$$L(\boldsymbol{x}, \boldsymbol{x}', y', \lambda) = \lambda \cdot \left(\hat{f}(\boldsymbol{x}') - y'\right)^2 + d(\boldsymbol{x}, \boldsymbol{x}')$$

The loss consists of the quadratic distance between the model prediction for the counterfactual x' and the desired outcome y' and the distance between the sample x to be explained and the counterfactual x'. λ is a parameter over which one maximises by iteratively solving for x' and increasing λ until a sufficiently close solution is found. The distance function d is the Manhattan distance weighted with the inverse median absolute deviation (MAD) of each feature j in feature set M.

$$d(\boldsymbol{x}, \boldsymbol{x}') = \sum_{j=1}^{M} \frac{\left|\boldsymbol{x}_j - \boldsymbol{x}'_j\right|}{MAD_j}$$

The total distance is the sum of all absolute differences of feature values between sample $\boldsymbol{x}$ and counterfactual $\boldsymbol{x}'$.

An algorithm for computing counterfactuals based on [140] is:

(1) Propose a sample $\boldsymbol{x}$ to be explained and the desired outcome y'.
(2) Sample a random sample as an initial counterfactual.
(3) Optimize the loss with the initially sampled counterfactual as starting point.
(4) While $\left|\hat{f}(\boldsymbol{x}') - y'\right| > \epsilon$:
   - Increase λ.
   - Optimize the loss with the current counterfactual as starting point.
   - Return the counterfactual that minimizes the loss.
(5) Repeat steps 2-4 and return the list of counterfactuals or the one that minimizes the loss.

The method does not handle categorical features with many different levels well. The authors of the method suggested running the method separately for each combination of feature values of the categorical features, but this will lead to a combinatorial explosion if you have multiple categorical features with many values. For example, six categorical features with ten unique levels would mean one million runs.

ABELE (Adversarial Black-box Explainer generating Latent Exemplars) targets image models by learning a latent space with an adversarial autoencoder, then using a genetic algorithm to explore neighbourhoods and learn a local surrogate (decision tree) that delivers both factual and counterfactual *exemplars* with simple rule paths [77]. Compared with purely boundary-hugging counterfactuals, ABELE's latent exemplars tend to be more realistic and visually interpretable (e.g., counterfactual retinal scans for a different diagnosis), though its reliance on image-specific latent spaces could make direct EHR application non-trivial.

One of the limitations of counterfactuals generally is their lack of uniqueness, i.e. multiple counterfactual explanations can be provided for the same prediction. The method above cannot efficiently handle categorical features with many different levels since it depends on running the method separately for each combination of feature values. An approach was proposed by [48] using four-objective loss that allows for multi-category counterfactual optimisation with a nondominated sorting genetic algorithm. Extended work on applications in healthcare like for glaucoma detection has shown better explainability than classical methods like saliency maps [33].

3.12 Large Language Models (LLMs)

Large language models and their autoregressive generation, size, and training opacity lead to separate challenges from conventional predictors [25]. Gradient- or perturbation-based techniques that work somewhat on smaller networks might not generalise to these methods [222]. In healthcare, their explainability problem consists of: (i) hallucinations producing convincing but false medical statements that can be hard to detect [5, 17]; (ii) opacity hiding training data composition and internal logic; and (iii) inconsistency arising when identical prompts give different outputs across runs or model versions [137].

Explainability Approaches. Work on LLM explainability for clinical tasks falls roughly into attention visualisation, chain-of-thought prompting, and mechanistic interpretability. Attention-based analyses trace weight patterns across transformer layers and heads, extending early BERT-era tools to modern architectures [176, 202]. In practice, though, the link between attention magnitude and causal importance is weak and unstable under small input changes [93]. Chain-of-thought prompting elicits intermediate reasoning steps and can improve task performance in some settings [86]. The difficulty is that these rationales are themselves model outputs. They inherit the same risks as the primary answers, including confident fabrication and explanations that do not track the computation used [192].

Commercial versus Open-Source Distinctions. What you can do with explainability depends heavily on whether the model is open or closed. Commercial APIs such as GPT, Claude, and Gemini usually expose only inputs and outputs, confining analysis to prompt engineering, behavioural testing, and post hoc audits. Open-source models such as LLaMA, Mistral, and OLMo grant access to gradients, attention, and training data documentation, which allows deeper study of error and bias patterns [76]. Openness does not immediately imply mitigation, however. Biomedical corpora making up a measurable fraction of training text does not correct the historical disparities in those sources. Hallucination and stochastic variance persist even with expert prompting and require guardrails, such as retrieval augmentation and constrained decoding. In reality, many clinical incidents could mix both a model fabricating a plausible recommendation and the user accepting it without cross-checking. Studies have shown that training clinicians on when and how to question model outputs can help use the tools more robustly [123, 187].

Cognitive Biases in Clinical LLMs. Framing effects can occur when equivalent clinical models give different assessments because the prompt emphasises a particular hypothesis. Anchoring bias occurs when early tokens, such as age or race, steer subsequent reasoning more than later laboratory findings, which can compound known disparities in clinical risk scoring [58, 154]. Availability effects favour common diagnoses and patterns over rare conditions, which is concerning for emergency triage and rare disease settings. These biases interact with traditional fairness issues and are harder to audit in commercial systems that do not show intermediate representations. Open-source transparency helps diagnose failure modes, but does not provide a path to remedy on its own.

Mitigation Strategies. Mitigation spans pre-processing, in-processing, and post-processing with trade-offs that depend on clinical context. Reweighting and stratified sampling can reduce representation gaps, but may remove a clinically meaningful signal. Constrained learning and adversarial debiasing embed parity objectives into training and can reduce disparities at some cost to overall accuracy. Post-processing with group-specific thresholds or rule edits guided by explanations preserves the base model and demands careful calibration and governance [113, 160]. For LLMs, mitigation must also address cognitive effects. Structured prompting can reduce framing sensitivity, ensembles, and self-consistency can temper anchoring on early cues, and retrieval augmentation can lift performance under rare conditions. None of these eliminates bias entirely, so the documentation of residual risks and monitoring plans is a standard yet insufficient part of deployment. Intersectional analysis should be routine, since the disparities at the intersections of age, gender, and race can be larger than along any single axis. Subgroup discovery with clinical oversight helps surface at-risk populations while avoiding over-fragmentation. Finally, decision curve analysis and calibration within groups help ensure that improvements in parity align with better outcomes for patients [201]. Because models degrade with shift, fairness drift should be monitored prospectively with alerts when performance gaps widen over time [133].

Worked examples of chain-of-thought prompting and retrieval-augmented explanation for clinical risk, together with a safety checklist, are provided in Listing S6 of the supplementary material. A stratified SHAP bias-analysis implementation, including demographic parity computation and a clinical interpretation checklist, is provided in Listing S7 of the supplementary material.

4 Data Modalities

Learning Objectives

(1) Match explainability methods to data characteristics
(2) Handle modality-specific challenges (e.g., temporal dependencies in EHR)
(3) Combine explanations across modalities
(4) Implement domain-appropriate evaluation protocols

Different clinical data modalities, imaging, text, genomics, and EHR, have different requirements for explainability. This section provides modality-specific guidance for method selection, implementation, and validation.

4.1 Imaging

Medical imaging was among the first successful applications of deep learning in healthcare, with the FDA clearing *Arterys* for cardiac MRI analysis in 2017 [103]. Beyond traditional radiology, image sources include endoscopic ultrasound, and increasingly, converted signal data like ECG spectrograms [32, 147]. While imaging applications span detection, segmentation, denoising, and report generation, the dominance of CNNs creates particular explainability challenges [12, 121, 142, 169].

While humans can visually inspect raw images, AI-learned patterns often go beyond conventional radiographic features [116, 124]. Images are represented as numeric matrices, but the patterns CNNs extract from these pixel values may not correspond to features radiologists traditionally examine. This disconnect makes explainability important for clinical trust and adoption.

Limitations include extensive labelling requirements, often requiring specialist annotations that create bottlenecks, leading to heavy reliance on data augmentation [42, 127]. Medical images typically exist in institutional silos, separated from complementary patient data and isolated across health systems. This fragmentation implies the need for multi-centre validation to ensure algorithms generalise beyond their training institutions [210].

4.2 Text

Clinical text, physician notes, radiology reports, discharge summaries, forms a substantial portion of electronic health records [31, 184]. Because textual reports can accompany other diagnostic modalities such as ECG recordings, echocardiography, and pathology, clinical text can be considered as part of multi-modal integration. In conditions with onset, such as sepsis, longitudinal clinical notes capture subtle temporal cues that isolated measurements cannot capture [71, 212].

NLP pipelines extract structured information from this unstructured text, whether through automated ICD coding or entity recognition from discharge summaries [120, 173]. However, clinical language poses particular challenges with pervasive jargon and abbreviations, complex negation patterns, and the sheer volume of free text that must be processed [184]. Annotation remains the main bottleneck, which has driven growing interest in unsupervised and semi-supervised learning approaches [131, 190].

Modern transformer architectures and large language models show promise for clinical tasks, including aspects of mental health triage and patient communication. Their opacity, however, raises domain-specific concerns. They are unable to detect non-verbal cues such as tone and body language, have the risk of hallucinating clinically harmful information, and carry well-documented biases inherited from training data [110]. These risks make explainability more important when such models are considered for clinical deployment.

4.3 Genomics

The structure of genomic data comes with multiple representational strategies. Nucleotide sequences (A, T, C, G) can be treated as text-like strings by RNNs, or converted into matrix representations for CNN-based analysis [8]. Applications span variant calling in Mendelian disorders through to complex trait prediction, and are increasingly combined with clinical data to map phenotype–genotype associations [97, 199, 207].

DeepVariant is an example the image-based approach which encodes read pileups at candidate variant sites as image tensors and classifying them with a convolutional neural network [111, 159]. While this representation allows genomic variant calling to benefit from advances in computer vision, it also distances the model's internal representations from the underlying biology and makes mechanistic interpretation more difficult.

Explainability in genomics has several challenges. One is motif redundancy, when a biologically important sequence pattern appears multiple times, individual instances become hard to distinguish from one another. Classic gradient-based saliency maps struggle here because saturated activations suppress gradient flow, while perturbation-based methods tend to undervalue redundant motifs since occluding any single copy leaves others to compensate [61]. Reference-based attribution methods such as DeepLIFT [102] and integrated gradients [189] partially address this by measuring contributions relative to a baseline sequence rather than relying on local gradients alone. Given these complexities, comparing multiple explainability approaches is necessary to be able to draw reliable conclusions in genomic settings.

4.4 Electronic Health Records

EHRs are a rich structured data source combining demographics, medical history, diagnoses, medications, vital signs, and laboratory results [3]. The COVID-19 pandemic showed their value at scale, when population-level EHR analyses directly informed public health decisions [211]. For risk prediction, EHRs offer longitudinal views that isolated images or point-in-time measurements cannot provide.

EHR data typically takes two primary forms: static or tabular features (demographics, comorbidities) and time-series measurements (vital signs, repeated laboratory tests). This heterogeneity usually requires flexible modelling approaches, from conventional regression for tabular data to recurrent architectures such as LSTMs and GRUs for temporal sequences [150]. Where unstructured clinical notes are available, NLP preprocessing is often needed to extract codified information [158].

Several practical challenges shape how EHR data can be used. In the United States, institutions generate records primarily for billing rather than research. And, often systematic missingness patters have to be handled carefully so as to not introduce systematic biases through assumptions of missingness-at-random or complete case analysis [209]. ICU settings have dense physiological monitoring, while primary care produces sparser but longer-term observations. Processing pipelines have to adapt with varying sampling rates, informative missingness (where absent laboratory tests may reflect clinical stability, discharge, or other non-random factors), and institution-specific documentation practices. Therefore, transparent reporting of preprocessing decisions, imputation strategies, approaches to irregularly sampled series, and bias mitigation methods should be considered when claiming reproducible research.

4.5 Multi-Modal Integration

Healthcare decisions naturally integrate multiple data streams: radiologists consider clinical history when interpreting images, and physicians combine laboratory trends with physical examination findings. Multi-modal AI seeks to mirror this synthesis, combining imaging with EHR data or genomics with clinical notes to improve both prediction and explanation [87].

Multi-modal explainability introduces different challenges, however. Models can develop unexpected dependencies, relying heavily on one modality while still incurring the computational cost of processing all inputs.

Explanation methods must represent each modality's contribution faithfully, guarding against bias toward visually interpretable channels such as imaging at the expense of more abstract features like laboratory values. Cross-modal attention mechanisms and modality-specific ablation studies offer ways to verify that explanations reflect the model's actual decision process rather than human preferences for certain data types [208].

In the Supplementary, we present an overview of selected papers across these modalities over the years (2019-2024) to illustrate overall patterns of XAI in clinical risk prediction.

5 Challenges and Future Research

5.1 Evaluation and Validation

It is important to note that many of the included methods are, in fact, practically limited. They are sometimes vulnerable to failures, redundant explanations, and wrong indications which makes interpretability on its own an insufficient attribute for achieving explainable and reliable AI clinical risk models [67]. Some work like [57, 81] and others have in recent years addressed the need for external validation if not as a companion to increased explainability then in place of it. These strategies for XAI in healthcare applications should be considered with caution as models externally validated on similar patient cohorts might not add additional trust or evaluate fairness more robustly than intuitive explainability methods. Ideally, a combination of both external validation with diverse stratified sub-population cohorts based on socio-economic and comorbidity groups and robust interpretability methods whether they be inherent or post-hoc is recommended to achieve explainability. An important extension should be to develop more robust explainability frameworks that can work across modalities and methodological contexts, as well as provide evidence for external multi-centre validation of proposed prediction models.

We suggest the adoption of more rigorous testing of applied explainability in clinical risk prediction modelling by testing the methods on synthetic datasets with known underlying generative factors. There are already some examples of such datasets extracted from existing and in-use clinical datasets like the Clinical Practice Research Datalink (CPRD) containing EHRs of millions of patients in the United Kingdom or UK Biobank which similarly contains a large amount of relatively complex and diverse patient data [82]. Testing that the implemented XAI methods correctly identify most of the known factors is an important reliability test before being proposed to be used in clinical decision-making systems. Demonstrating that XAI methods recover known factors strengthens the case for clinical adoption. In our collection of EHR time-series applications, we highlighted recent developments in implementing quantitative evaluations using a combination of corruption and ranking solutions based on adapted AUROC and F1 scores. The field still lacks robust frameworks for evaluating and comparing explainability across EHR and other clinical risk prediction domains.

5.2 Clinical Evaluation Methodologies in XAI

Building on our taxonomy (Section 3), we systematically characterise clinical evaluation approaches. Our analysis summarises practices into five methodological approaches, codified through the ClinEval and EvalSetup schema.

We use a consistent vocabulary for evaluation across the paper. *Expert review (ER)* refers to structured assessment by qualified clinicians who inspect model outputs or explanations against a predefined rubric, with reliability quantified where possible (e.g., Cohen's κ, ICC with 95% CIs). *Alignment with knowledge (AK)* tests whether explanations accord with clinical knowledge defined *a priori*, guidelines, ontologies, or curated literature, and should report explicit overlap or uncertainty and not just narrative agreement. Under *decision impact (DI)*, evaluators measure how explanations change clinician behaviour in controlled settings where relevant endpoints include diagnostic accuracy, triage time, confidence, and appropriate reliance (following the system when correct, overriding it when wrong), summarised by over- and under-reliance rates and reliance calibration curves. *Patient-facing (PF)* evaluation of comprehensibility and usefulness for patients or carers through instruments measuring comprehension, decisional conflict, trust, and usability. Finally, *workflow integration (WI)* captures effects during routine use, usage patterns, safety signals, and, when feasible, downstream outcomes such as adherence, readmission, or time to treatment. Throughout, calibration denotes agreement between predicted risks and observed outcomes within relevant groups, reported with slope, intercept, and visual checks.

Expert review establishes face validity, do explanations align with clinical reasoning? Appropriate reliance metrics try to gauge calibrated trust, do clinicians know when to follow versus override AI recommendations? Clinical impact studies then test whether the system actually improves patient care. Trust, though, is contextual rather than binary. A clinician may trust a system for routine cases yet remain sceptical when the presentation is complex. Our tiered evaluation framework, introduced in the next section, accounts for this by requiring stratified assessment across case difficulty, expertise levels, and clinical contexts. Unchecked trust leads to automation bias; unchecked distrust leads to underutilisation. Effective XAI should steer clinicians toward appropriate, evidence-based reliance [92].

5.3 Clinical Evaluation Standards

Drawing on our analysis, we propose a Minimum Clinical Validation Standard. Any study claiming clinical relevance should recruit at least three independent reviewers with documented domain expertise, specify evaluation protocols and blinding procedures in advance, and report inter-rater reliability (e.g., κ indicating at least substantial agreement). Benchmarking against established clinical baselines can help close the loop and make validation reproducible across institutions. Dataset choice matters as well, a small number of public benchmarks dominate the literature, while many studies rely on custom panels that trade experimental control for clinical realism, but rarely release sufficient detail for replication.

Our three-tier framework links clinical claims to appropriate evaluation standards. We used recent critical appraisals of performance metrics and fairness in clinical AI to help inform the design of each tier [133, 195].

(1) **Tier 1 (Basic Clinical Review).** Evaluators combine expert review with alignment analysis, reporting inter-rater statistics, uncertainty estimates, and group-wise performance. We recommend pairing AUROC with calibration assessment, since discrimination alone can obscure clinically dangerous miscalibration. Stratified calibration plots serve as a practical tool and identify performance differences across sensitive attributes at the earliest auditing stage [133].
(2) **Tier 2 (Decision Impact).** Controlled experiments can quantify how explanations shape decision quality and appropriate reliance, stratified by user expertise and case difficulty. This tier treats XAI as human–AI teaming with interfaces adapting to the clinician instead of simply delivering uniform explanations regardless of context [187]. Clinical utility metrics, such as subgroup net benefit capture decision quality at meaningful thresholds [133, 201].
(3) **Tier 3 (Clinical Integration).** Prospective deployment requires workflow measurement and outcome tracking over time. A problem with fairness drift where disadvantaged groups often experience performance degradation earlier than others as data distributions shift post-deployment can be addressed by metrics connecting fairness to clinical utility, as predictive AI that fails to improve patient outcomes is of limited use in routine care [133].

Sample size guidance for fairness evaluation barely exists, despite the vast imbalances between demographic groups in most clinical datasets. Confidence intervals around fairness metrics are rarely reported, and intersectional analyses remain largely infeasible at current sample sizes. Future work should address these shortcomings through uncertainty quantification and systematic real-world evaluation [1, 133]. On the governance side, accountability frameworks need to include technical dimensions such as version control and audit trails, clinical dimensions that preserve human oversight and clinician responsibility, and organisational dimensions including governance committees and incident protocols. Legal liability for adverse outcomes involving AI remains unresolved, developers, deployers, and clinicians each bear some responsibility, but it is not always clearly apportioned [148, 162, 164, 167].

TRIPOD+AI offers a practical template example for reporting rigour where studies should state their evaluation objectives, describe reviewer characteristics, provide complete instruments, report metrics with confidence intervals, and share materials for replication [45]. Defining fairness itself, however, requires interdisciplinary dialogue among clinicians, patients, ethicists, and affected communities. Unilateral definitions imposed by unrepresentative groups are not acceptable, and the evaluation of an explanation deserves the same rigour we demand of the predictive model [133].

5.4 Privacy-Preserving Explainability

Explanations enhance transparency and trust, but they can also leak sensitive patient information or invite adversarial exploitation [180]. The tension matters because regulations like GDPR and HIPAA impose strict data protection requirements, even as clinical governance increasingly demands interpretable decisions [163].

Example-based explanations risk revealing training instances, which leads to membership inference, determining whether a specific patient appeared in the training data [180]. Feature-based explanations are not immune: Shokri et al. showed that backpropagation-based explanations leak significant information about individual training records through membership inference [179]. More recently, Luo et al. demonstrated that adversaries can reconstruct private input features from Shapley value explanation reports, even with limited query access to the model [128]. In clinical contexts, these vulnerabilities could enable patient reidentification, gaming of risk stratification scores, or discriminatory use of inferred health information [163].

Differential privacy (DP) offers some protection by adding calibrated noise to explanations, though the resulting noise–utility trade-off can be severe [156]. Distributed learning has prompted several responses. DC-SHAP produces consistent explanations across participants in privacy-preserving distributed learning without exchanging raw data [26]. Bonsignori et al. adapt FastSHAP with differential privacy guarantees for federated learning and show that explanation fidelity can be maintained under reasonable privacy budgets [27]. Naretto et al. offer practical guidance by empirically evaluating the privacy exposure of different XAI methods. Their findings suggest that global explainers based on interpretable surrogates leak substantially more information than local alternatives, which has direct implications for method selection in clinical deployments [149]. Corcuera Bárcena et al. show that inherently interpretable model families, specifically federated fuzzy regression trees, can preserve privacy through federation without sacrificing transparency, evidence that the two goals need not conflict [46]. Neel et al. survey the landscape of privacy attacks on model explanations and review countermeasures, from noise injection and secure multi-party computation to federated explanation generation [153].

The GDPR's supposed "right to explanation" should be interpreted conservatively. Wachter et al. showed that the regulation mandates only limited information about "the logic involved" in automated decision-making, falling well short of requiring comprehensive explanations [204]. This gap means privacy-preserving methods can still convey useful information without exposing the full decision logic. Healthcare organisations face a tension between these transparency requirements and the GDPR's own privacy mandates: Article 22 calls for "suitable safeguards" in automated decision-making, but these need not take the form of detailed explanations that compromise patient confidentiality [204]. Before deciding how much explanatory detail to expose, practitioners should consider how sensitive the features are, whether patient trajectories are unique enough to be reidentifiable, and what reidentification risks remain [163]. There is no universal answer here. The right balance will vary with the regulatory environment and institutional risk appetite, and ultimately with what is at stake clinically.

6 Conclusion

Our tutorial presents a curated rather than exhaustive view of the literature. Explaining and reliably deploying ML models in clinical settings demands a combination of technical innovation, external validation, human-centred design, and cross-disciplinary collaboration spanning ethics, law, and clinical practice. Predictive AI can reduce costs and backlogs while improving patient care, but it is not a panacea, human expertise and domain knowledge remain indispensable. Previous attempts at clinical XAI make one lesson clear: in healthcare, trust, interpretability, and fairness matter as much as predictive performance, and no single discipline can deliver all of them alone.

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